

Exercice : 01

$$f(x,y) = \sqrt{x^2 + y^2}$$

$$\frac{df}{dx} = \frac{2x}{2\sqrt{x^2+y^2}} = \frac{x}{\sqrt{x^2+y^2}}$$

$$\frac{df}{dy} = \frac{2y}{2\sqrt{x^2+y^2}} = \frac{y}{\sqrt{x^2+y^2}}$$

$$2 \quad f(x,y) = x \sin(x+y)$$

$$\frac{df}{dx}(x,y) = \sin(x+y) + x \cos(x+y)$$

$$\frac{df}{dy}(x,y) = x \cos(x+y)$$

$$[3] \quad f(x,y) = y^5 - 3xy$$

$$\frac{df}{dx} = -3y$$

$$\frac{df}{dy} = 5y^4 - 3x$$

$$[4] \quad f(x,y) = x^2 \cos(e^{xy})$$

$$\frac{df}{dx} = 2x \cos(e^{xy}) - x^2 y e^{xy} \sin(e^{xy})$$

$$\frac{df}{dy} = -x^2 e^{xy} \sin(e^{xy})$$

Exercice : 02

$$f: \mathbb{R}^2 \rightarrow \mathbb{R} \quad \text{et} \quad g: \mathbb{R} \rightarrow \mathbb{R}$$

$$g(t) = f(2t, 1+t^2)$$

$$g(t) = f(u(t), v(t))$$

$$g'(t) = u'(t) \frac{df}{dx}(u(t), v(t)) + v'(t) \frac{df}{dy}(u(t), v(t))$$

$$\text{On a : } u(t) = 2t \quad \text{et} \quad v(t) = 1+t^2$$

$$\text{Alors } u'(t) = 2 \quad \text{et} \quad v'(t) = 2t$$

$$\text{Donc } g'(t) = 2 \frac{df}{dx}(2t, 1+t^2) + 2t \frac{df}{dy}(2t, 1+t^2)$$

Exercice : 03

Calculer des dérivées partielles d'ordre.

$$1) \quad f(x,y) = x^2(x+y) = x^3 + x^2y$$

$$\frac{df}{dx}(x,y) = 3x^2 + 2xy$$

$$\frac{df}{dy}(x,y) = x^2$$

$$\frac{d^2f}{dx^2}(x,y) = 6x + 2y, \quad \frac{d^2f}{dy^2}(x,y) = 0$$

$$\frac{d^2f}{dy dx}(x,y) = 2x \quad \text{et} \quad \frac{d^2f}{dx dy}(x,y) = 2x$$

Exercice: 04

$$f(x,y) = \begin{cases} \frac{xy^3}{x^2+y^2} & \text{si } (x,y) \neq (0,0) \\ 0 & \text{si } (x,y) = (0,0) \end{cases}$$

1) Montrer que f est de classe C^1 sur $\mathbb{R}^2$

$$\forall (x,y) \in \mathbb{R}^2 \setminus (0,0), f(x,y) = \frac{xy^3}{x^2+y^2}$$

$$\frac{df}{dx}(x,y) = \frac{y^3(x^2+y^2) - 2x(xy^3)}{(x^2+y^2)^2}$$

$$= \frac{y^5 - x^2y^3}{(x^2+y^2)^2}$$

$$\frac{df}{dx}(0,0) = \lim_{t \rightarrow 0} \frac{f(t,0) - f(0,0)}{t} = 0$$

$$\frac{df}{dy}(0,0) = \lim_{t \rightarrow 0} \frac{f(0,t) - f(0,0)}{t} = 0$$

$$\frac{df}{dx}(x,y) = \begin{cases} \frac{y^5 - x^2y^3}{(x^2+y^2)^2} & \text{si } (x,y) \neq (0,0) \\ 0 & \text{si } (x,y) = (0,0) \end{cases}$$

$\forall (x,y) \in \mathbb{R}^2 \setminus (0,0)$ and

$$\begin{aligned} \frac{df}{dy}(x,y) &= \frac{3xy^2(x^2+y^2) - 2y(xy^3)}{(x^2+y^2)^2} \\ &= \frac{3x^3y^2 + xy^4}{(x^2+y^2)^2} \end{aligned}$$

$$\frac{df}{dy}(0,0) = \lim_{t \rightarrow 0} \frac{f(0,t) - f(0,0)}{t} = 0$$

$$\text{Donc } \frac{df}{dy}(x,y) = \begin{cases} \frac{3x^3y^2 + xy^4}{(x^2+y^2)^2} & \text{si } (x,y) \neq (0,0) \\ 0 & \text{si } (x,y) = (0,0) \end{cases}$$

$$\forall (x,y) \in \mathbb{R}^2 \setminus (0,0), \frac{df}{dx}(x,y) = \frac{y^5 - x^2y^3}{(x^2+y^2)^2}$$

c'est une fonction rationnelle qui est continue sur $\mathbb{R}^2 \setminus (0,0)$.

$$\left| \frac{df}{dx}(x,y) \right| = \left| \frac{y^5 - x^2y^3}{(x^2+y^2)^2} \right|$$

$$\leq \frac{|y|^5}{(x^2+y^2)^2} + \frac{x^2|y|^3}{(x^2+y^2)^2}$$

$$\leq \frac{|y| \times y^4}{(x^2+y^2)^2} + \frac{|y| \times x^2 y^2}{(x^2+y^2)^2}$$

$$\leq |y| + \frac{|y|}{4} = \frac{5|y|}{4}$$

$$\text{donc } \lim_{(x,y) \rightarrow (0,0)} \frac{dg(x,y)}{dx} = 0 = \frac{dg(0,0)}{dx}$$

Alors $\frac{dg}{dx}$ est continu sur $\mathbb{R}^2$

$$y^2 \leq y^2 + x^2$$

$$(y^2)^2 \leq (x^2 + y^2)^2$$

$$\frac{|y| \times y^4}{(x^2 + y^2)^2} \leq \frac{|y| \times y^4}{y^4} = |y|$$

$$(|x| - |y|)^2 = x^2 + y^2 - 2|y||x| \geq 0$$

$$(x^2 + y^2)^2 \geq (2|x||y|)^2$$

$$\frac{|y|(x^2 + y^2)^2}{4(x^2 + y^2)^2} \geq \frac{4x^2 y^2 |y|}{4(x^2 + y^2)^2}$$

Exercice 105

$$1 \quad f(x,y) = x^2 + xy + y^2 - 3x - 6y$$

Dérivées partielles.

$$\begin{cases} \frac{df}{dx}(x,y) = 2x + y - 3 \\ \frac{df}{dy}(x,y) = x + 2y - 6 \end{cases}$$

Point Critique

$$\begin{cases} \frac{df}{dx}(x,y) = 0 \\ \frac{df}{dy}(x,y) = 0 \end{cases} \Rightarrow \begin{cases} 2x + y - 3 = 0 \quad (1) \\ x + 2y - 6 = 0 \quad (2) \end{cases}$$

$$\Rightarrow \begin{cases} 2x + y - 3 = 0 \\ -2x - 4y + 12 = 0 \end{cases} \Rightarrow \begin{cases} -3y + 9 = 0 \\ x = 0 \end{cases} \Rightarrow \begin{cases} y = 3 \\ x = 0 \end{cases}$$

donc (0,3) est un point Critique

$$f(x,y) - f(0,3) = x^2 + xy + y^2 - 3x - 6y + 9$$

$$= x^2 + x(y-3) + (y-3)^2$$

$$= \left(x + \frac{1}{2}(y-3)\right)^2 - \frac{1}{4}(y-3)^2 + (y-3)^2 \geq 0$$

Alors $\forall x,y \in \mathbb{R}^2, f(x,y) \geq f(0,3)$

Donc (0,3) est un minimum

$$2 \quad f(x,y) = x^2 + 2y^2 - 2xy - 2x + 5$$

Dérivées partielles

$$\frac{df}{dx}(x,y) = 2x - 2y$$

Et

$$\frac{df}{dy}(x,y) = 4y - 2x - 2$$

Points Critique.

$$\begin{cases} \frac{df}{dx}(x,y) = 0 \\ \frac{df}{dy}(x,y) = 0 \end{cases} \Rightarrow \begin{cases} 2x - 2y = 0 \Rightarrow x = y \\ 4y - 2x - 2 = 0 \\ 2x - 2 = 0 \Rightarrow y = 1 \\ x = 1 \end{cases}$$

Alors (1,1) est un point Critique.

$$\begin{aligned} f(x,y) - f(1,1) &= x^2 + 2y^2 - 2xy - 2x + 5 \\ &= x^2 - 2xy + y^2 + y^2 - 2x + 1 \\ &= (x-y)^2 + (y-1)^2 \geq 0 \end{aligned}$$

donc (1,1) est un minimum

3

$$f(x,y) = x^3 + y^3$$

$$\begin{cases} \frac{df}{dx}(x,y) = 3x^2 \\ \frac{df}{dy}(x,y) = 3y^2 \end{cases}$$

Point Critique

$$\frac{df}{dx} = 0 \quad \text{et} \quad \frac{df}{dy}(x,y) = 0$$

$$\Rightarrow x=0 \quad \text{et} \quad y=0$$

(0,0) est un point Critique.

Soit $n \in \mathbb{N}^*$

$$f(n,0) = n^3 > 0 = f(0,0)$$

donc (0,0) n'est pas un maximum

$$f(-n,0) = -n^3 < 0 = f(0,0)$$

donc (0,0) n'est minimum par
Suite (0,0) n'est ni minimum
ni maximum

Exercice 106

$$Z(t) = f(x(t), y(t))$$

$$f(x,y) = x^2 + y^2 + xy, \quad x(t) = \sin(t), \quad y(t) = e^t$$

$$Z'(t) = \dot{x}(t) \frac{df}{dx}$$